

Borghetto 2024

Survey by A. Rovere and E. Casella

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Survey Data

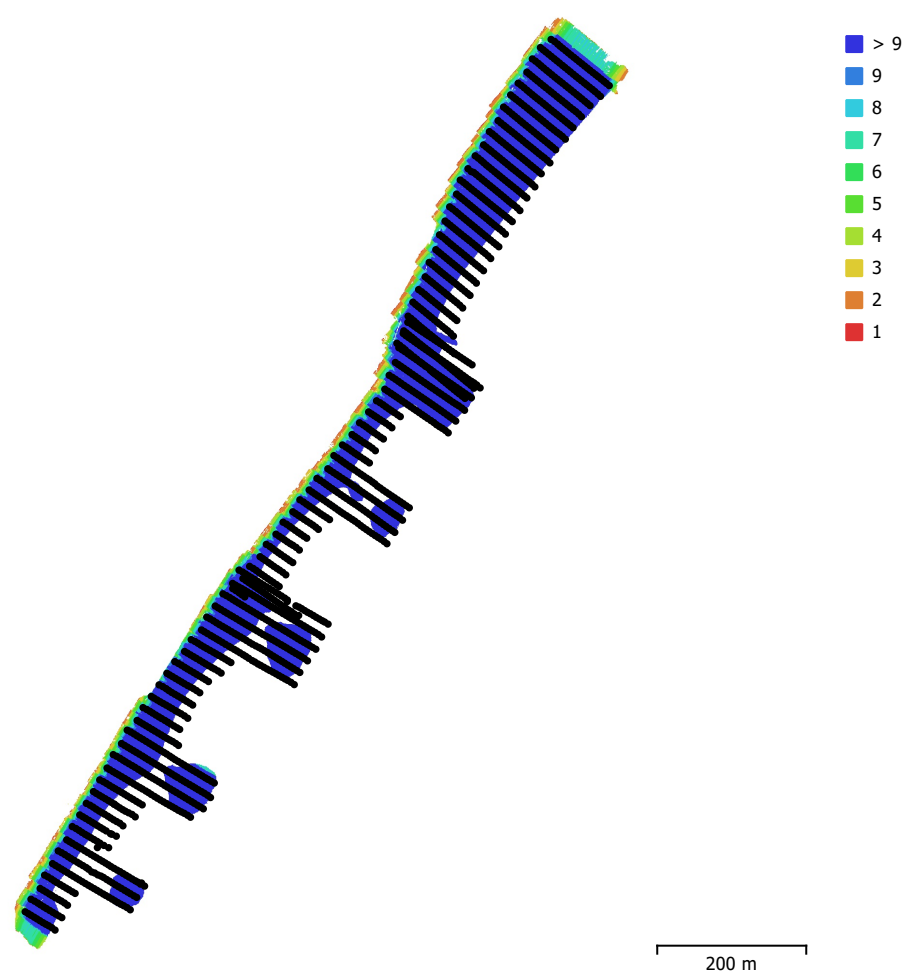


Fig. 1. Camera locations and image overlap.

Number of images:	1,502	Camera stations:	1,502
Flying altitude:	47.1 m	Tie points:	1,978,482
Ground resolution:	1.45 cm/pix	Projections:	10,199,274
Coverage area:	0.104 km ²	Reprojection error:	1.23 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
FC7303 (4.49mm)	4000 x 2250	4.49 mm	1.76 x 1.76 μm	No

Table 1. Cameras.

Camera Calibration

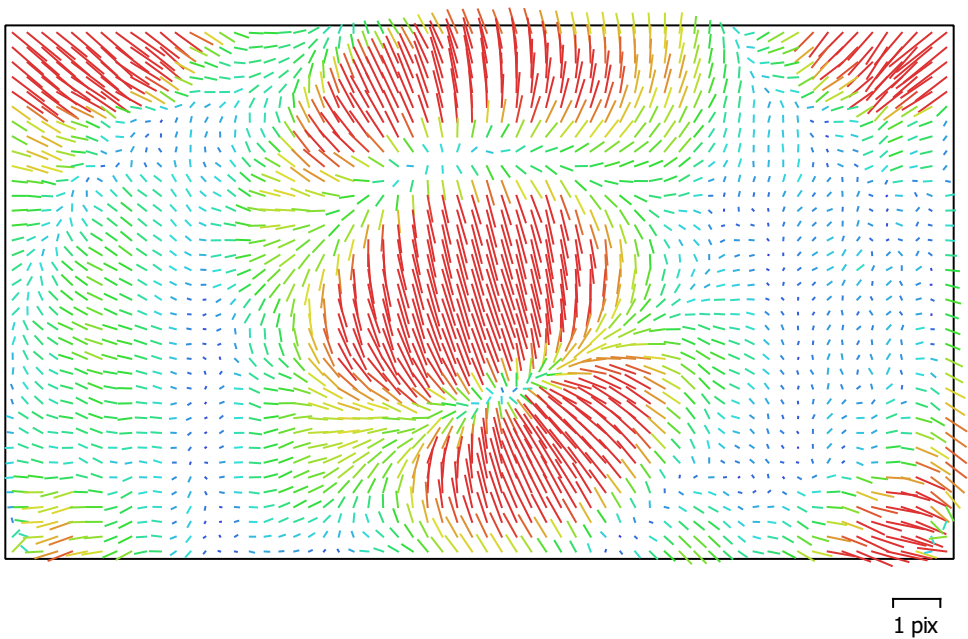


Fig. 2. Image residuals for FC7303 (4.49mm).

FC7303 (4.49mm)

1502 images

Type	Resolution	Focal Length	Pixel Size
Frame	4000 x 2250	4.49 mm	1.76 x 1.76 μ m

	Value	Error	F	Cx	Cy	K1	K2	K3	K4
F	3129.33	0.2	1.00	-0.55	-0.85	0.08	-0.19	0.24	-0.28
Cx	-27.7405	0.03		1.00	0.47	-0.03	0.09	-0.12	0.14
Cy	-67.1012	0.04			1.00	-0.08	0.17	-0.21	0.25
K1	0.0422349	4.6e-05				1.00	-0.97	0.91	-0.85
K2	-0.307107	0.00034					1.00	-0.98	0.95
K3	0.73168	0.00098						1.00	-0.99
K4	-0.610661	0.00094							1.00

Table 2. Calibration coefficients and correlation matrix.

Ground Control Points

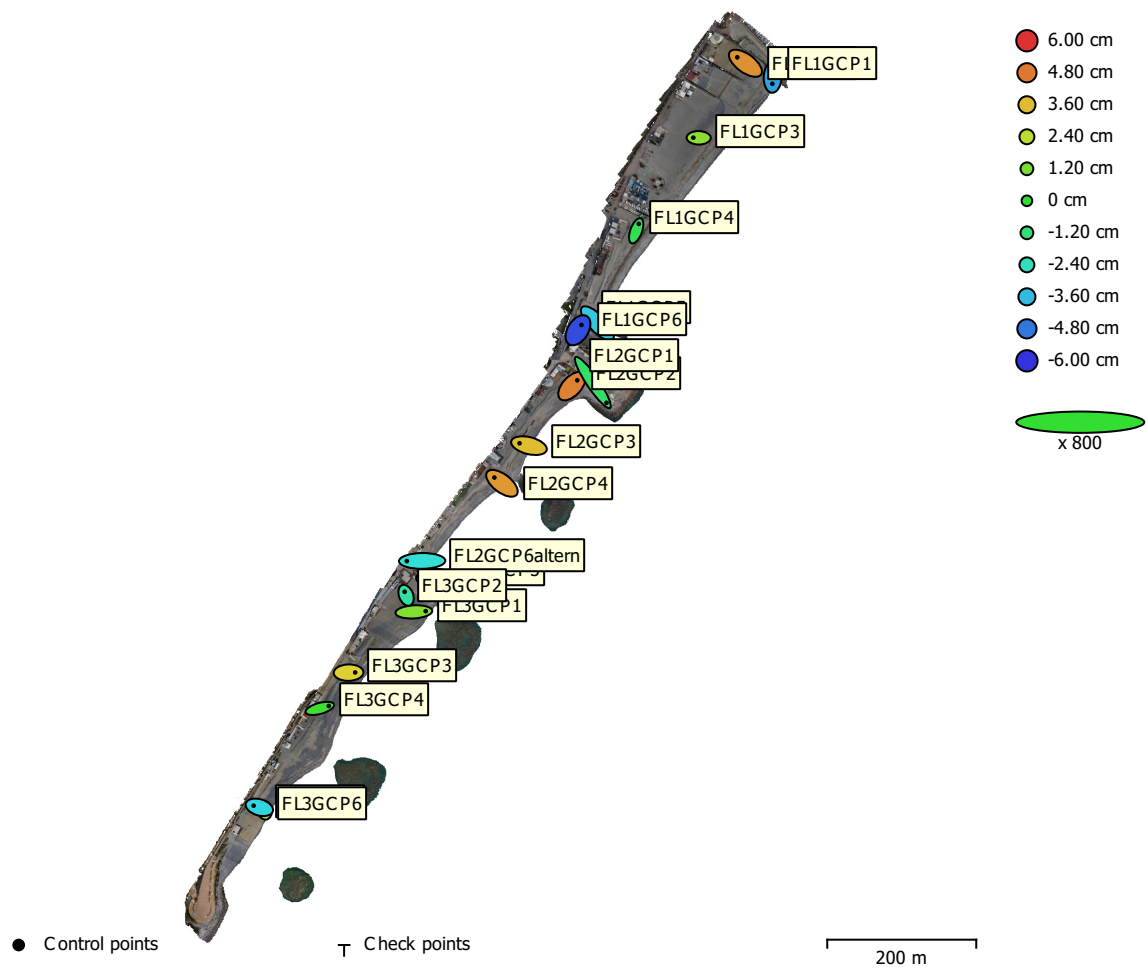


Fig. 3. GCP locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.

Estimated GCP locations are marked with a dot or crossing.

Count	X error (cm)	Y error (cm)	Z error (cm)	XY error (cm)	Total (cm)
18	2.72315	2.19916	3.14633	3.50026	4.70651

Table 3. Control points RMSE.

X - Easting, Y - Northing, Z - Altitude.

Label	X error (cm)	Y error (cm)	Z error (cm)	Total (cm)	Image (pix)
FL2GCP2	1.7925	1.94458	4.6188	5.32238	1.029 (19)
FL2GCP1	4.60195	-6.80575	-0.943549	8.26961	0.784 (19)
FL2GCP3	-3.17418	0.794937	3.56565	4.83955	0.695 (16)
FL2GCP4	-2.54075	1.9394	4.20165	5.27925	0.909 (17)
FL2GCP5	-3.07546	-0.374004	-2.64347	4.07262	0.785 (17)
FL2GCP6altern	-5.1357	-0.059319	-2.87353	5.88525	1.481 (6)
FL1GCP2	-2.73977	1.98763	4.34909	5.51104	0.897 (21)
FL1GCP1	-0.0926436	-2.24644	-4.08823	4.6657	1.022 (14)
FL1GCP3	-1.79896	0.0279443	1.56571	2.38505	0.823 (20)
FL1GCP4	0.872173	2.33712	-0.592259	2.56391	1.459 (21)
FL1GCP5	3.22614	-3.41446	-3.35415	5.77208	0.681 (20)
FL1GCP6	1.15035	1.79309	-5.52974	5.92591	1.405 (24)
FL3GCP1	3.95892	0.238273	1.40199	4.20659	0.933 (18)
FL3GCP2	-0.475695	1.18984	-1.94346	2.32789	0.856 (18)
FL3GCP3	2.21693	0.056466	3.24739	3.93237	0.899 (19)
FL3GCP4	2.99376	0.821868	0.200864	3.11102	0.948 (14)
FL3GCP5	0.139911	-0.78864	1.96463	2.12162	0.881 (20)
FL3GCP6	-1.91948	0.557456	-3.14737	3.72842	0.825 (12)
Total	2.72315	2.19916	3.14633	4.70651	0.983

Table 4. Control points.
X - Easting, Y - Northing, Z - Altitude.

Digital Elevation Model

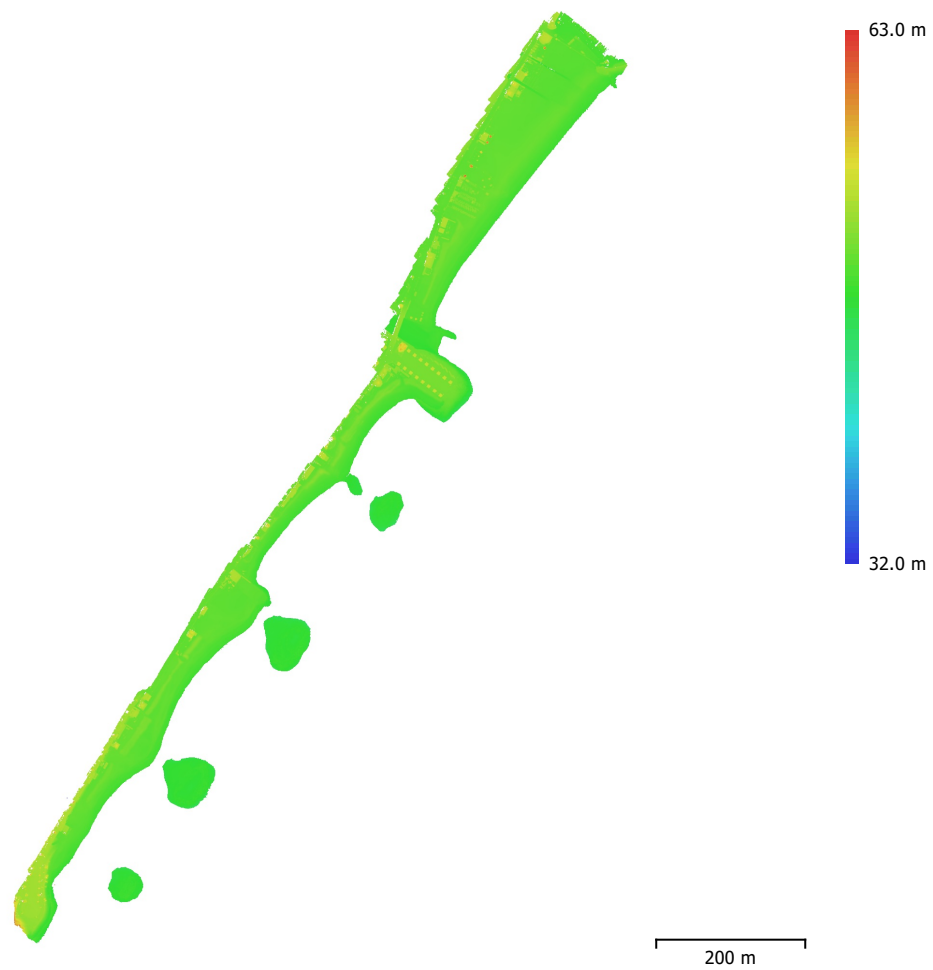


Fig. 4. Reconstructed digital elevation model.

Resolution: 2.90 cm/pix
Point density: 0.119 points/cm²

Processing Parameters

General

Images	1502
Aligned images	1502
Markers	18
Coordinate system	WGS 84 / UTM zone 32N (EPSG::32632)
Rotation angles	Yaw, Pitch, Roll

Shapes

Coordinate system	WGS 84 / UTM zone 32N (EPSG::32632)
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Tie Points

Points	1,978,482 of 2,173,166
Point colors	3 bands, uint8
RMS reprojection error	0.279556 (1.22779 pix)
Max reprojection error	1.58636 (65.5126 pix)
Mean key point size	4.22901 pix
Key points	No
Average tie point multiplicity	5.51984

Alignment parameters

Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	10,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	No
Matching time	5 minutes 57 seconds
Matching memory usage	1.75 GB
Alignment time	20 minutes 27 seconds
Alignment memory usage	1.54 GB

Optimization parameters

Parameters	f, cx, cy, k1-k4
Adaptive camera model fitting	No
Exclude corners	No
Optimization time	36 seconds
Date created	2025:03:27 16:04:19
Software version	2.1.3.18946
File size	234.74 MB

Point Cloud

Points	148,959,697
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Point attributes

Color	3 bands, uint8
Normal	

Point classes

Created (never classified)	148,959,697
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Depth maps generation parameters

Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	22 minutes 51 seconds
Memory usage	3.03 GB

Point cloud generation parameters

Source data	Depth maps
Processing time	53 minutes 31 seconds

Memory usage	5.48 GB
Date created	2025:04:24 11:22:10
Software version	2.1.3.18946
File size	2.29 GB
Model	
Faces	34,663,245
Vertices	17,354,243
Vertex colors	3 bands, uint8
Depth maps generation parameters	
Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	22 minutes 51 seconds
Memory usage	3.03 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	1 hours 31 minutes
Memory usage	5.21 GB
Reconstruction parameters	
Surface type	Height field
Source data	Point cloud
Interpolation	Enabled
Strict volumetric masks	No
Processing time	8 minutes 36 seconds
Memory usage	7.45 GB
Date created	2025:04:29 17:01:33
Software version	2.1.3.18946
File size	678.00 MB
DEM	
Size	28,371 x 43,033
Resolution	2.90 cm/pix
Coordinate system	WGS 84 / UTM zone 32N (EPSG::32632)
Reconstruction parameters	
Source data	Point cloud
Interpolation	Disabled
Processing time	1 minutes 2 seconds
Memory usage	308.34 MB
Date created	2025:10:29 11:43:31
Software version	2.2.2.21069
File size	462.36 MB
Orthomosaic	
Size	9,971 x 13,277
Resolution	10.0 cm/pix
Coordinate system	WGS 84 / UTM zone 32N (EPSG::32632)
Colors	3 bands, uint8
Orthophotos	6.31 MB
Reconstruction parameters	
Blending mode	Mosaic
Surface	Point cloud
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	14 seconds
Memory usage	784.04 MB
Date created	2025:10:29 11:55:16
Software version	2.2.2.21069
File size	30.72 MB
System	
Software name	Agisoft Metashape Professional
Software version	2.2.2 build 21069
OS	Windows 64 bit

RAM	127.82 GB
CPU	13th Gen Intel(R) Core(TM) i7-13700F
GPU(s)	NVIDIA GeForce RTX 4070